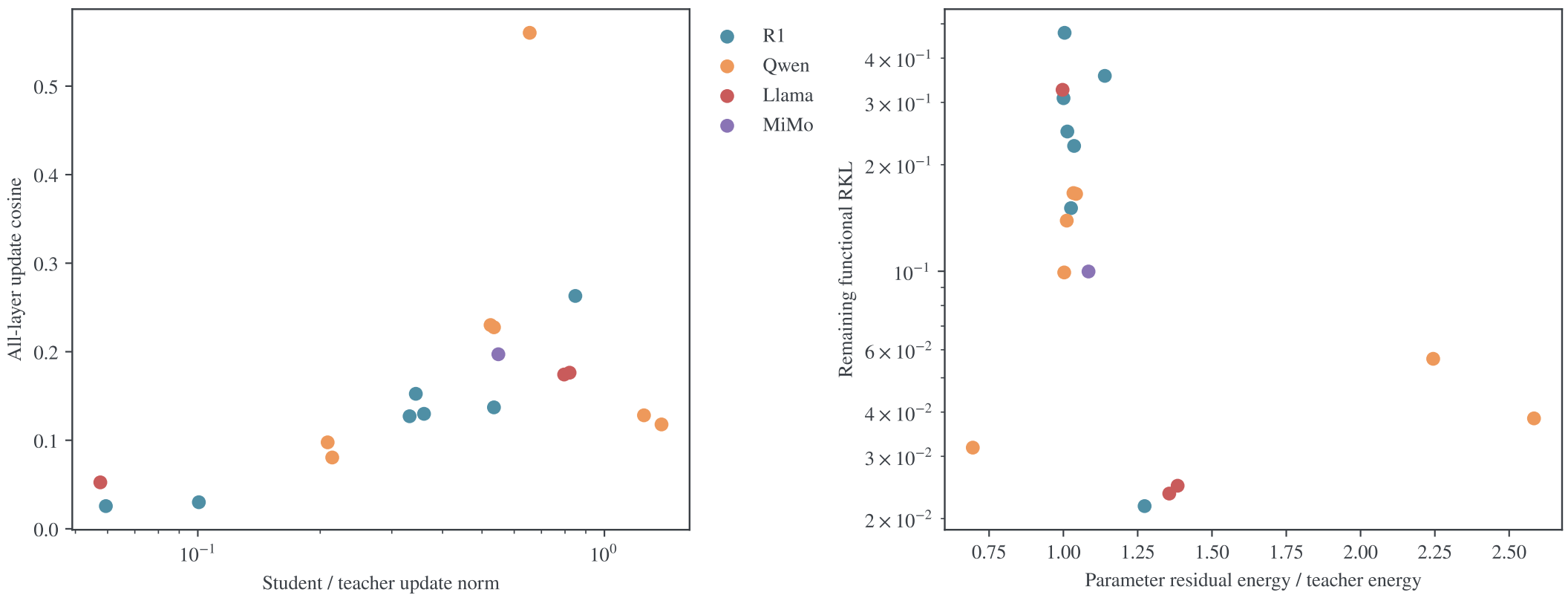
All-layer update cosine
0.0
0.1
0.2
0.3
0.4
0.5
10^{-1}
10^{0}
Student / teacher update norm
R1
Qwen
Llama
MiMo
Remaining functional RKL
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
0.75
1.00
1.25
1.50
1.75
2.00
2.25
2.50
Parameter residual energy / teacher energy
All-layer energy aggregation; model endpoints describe learning paths